



Modeling Information Diffusion on Twitter Using BFS

A case study of the SNAP Higgs Twitter Dataset — exploring how scientific news spreads through a large social network using Breadth-First Search.

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AI statement: AI used in generating pictures in the slides

Introduction & Research Question



This project models the Twitter retweet network as a **directed graph** and applies **Breadth-First Search (BFS)** to analyze information diffusion during a major scientific event.

 Core Question: Is diffusion potential better explained by **early appearance**, **random structural position**, or **stronger directed structural position**?

Dataset & Graph Construction

Built from the SNAP Higgs Twitter Dataset — retweet network + activity log. Each node = user; each directed edge = retweet. The graph is **reversed** before BFS to model upstream information flow.

256K

Nodes

Users in the graph

328K

Edges

Retweet relations

87.27%

Largest WCC

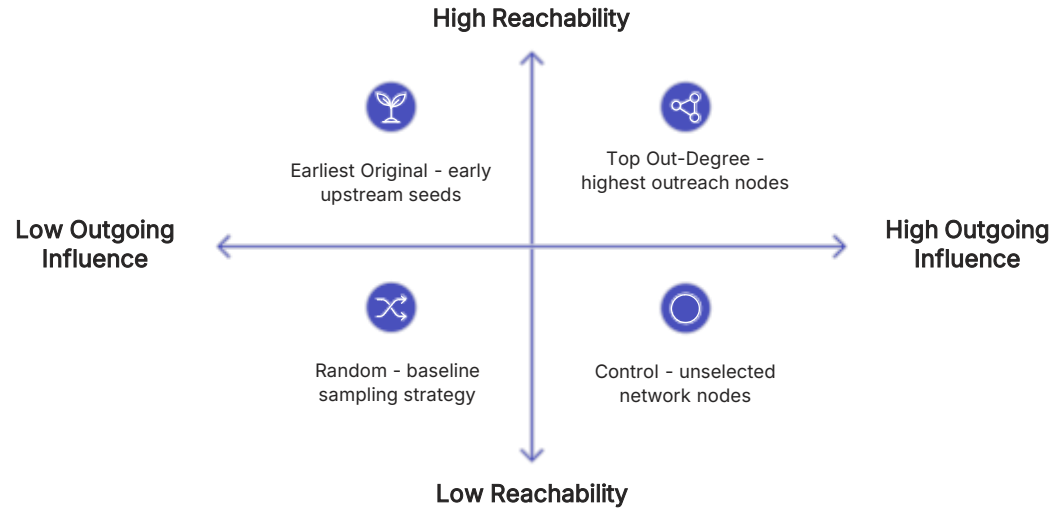
Weakly connected

0.38%

Largest SCC

Strongly connected

Methodology: BFS & Source Strategies



Metrics Computed Per Source

- Reachable nodes & reachable ratio
- Maximum BFS depth
- Average & median shortest-hop distance
- WCC / SCC membership
- Out-degree vs. diffusion reach correlation

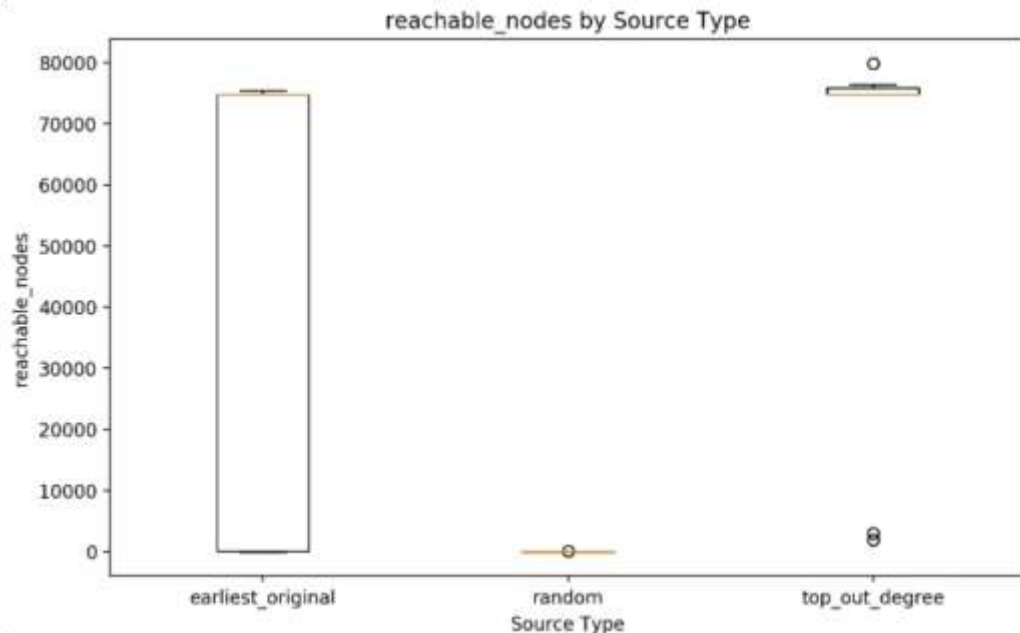
BFS explores the graph level by level, matching the layered structure of information diffusion.



Global Graph Structure

The large WCC shows most nodes share a broad region — but the tiny SCC (0.38%) reveals the graph is **not mutually reachable** once direction is preserved. The reversed retweet graph behaves as a **highly directional diffusion structure**, not a bidirectional discussion network.

Comparing Source Groups: Reach



Random

Mean reach **~1.2 nodes**. Being in a large WCC does not imply outward diffusion power.

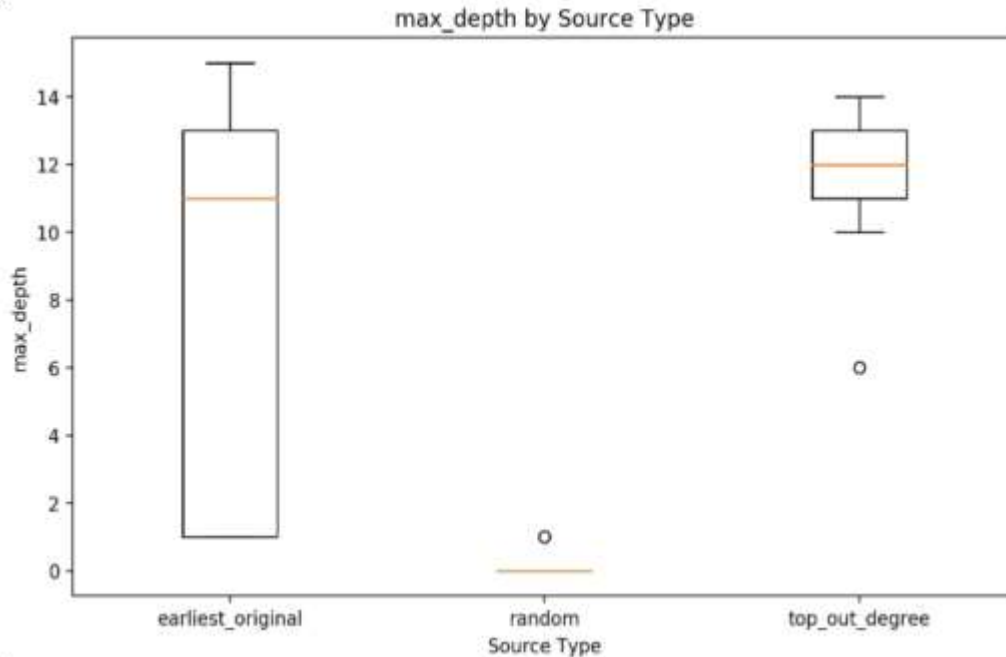
Earliest Original

Mean reach **~44,919 nodes**, but highly heterogeneous. Early appearance alone is insufficient.

Top Out-Degree

Mean reach **~61,092 nodes**. Median max depth 12. Strongest group overall.

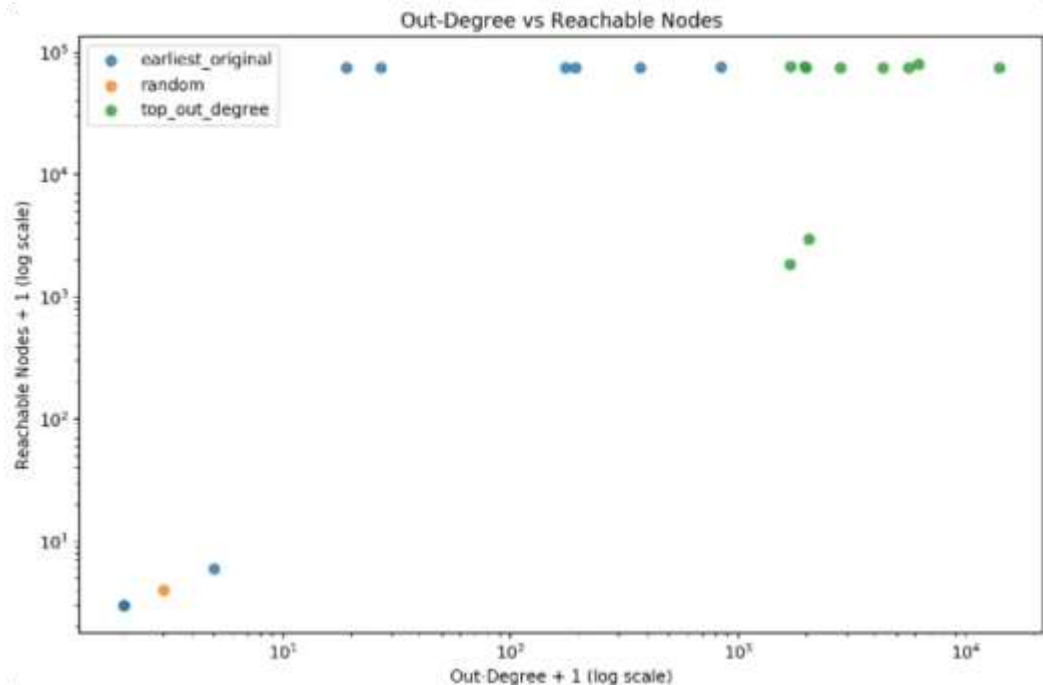
Comparing Source Groups: Max Depth



The max depth chart reinforces the reach findings. **Top out-degree** sources achieve consistently deep BFS trees (median depth 12), while **random** nodes are nearly flat — most reach depth 0 or 1.

📌 **50%** of top out-degree users belong to the largest SCC, confirming their strong directed structural position.

Degree–Reach Relationship



Overall Correlations

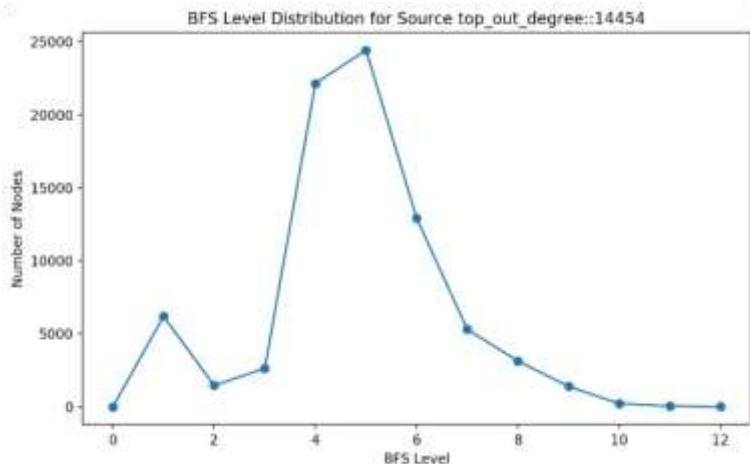
Spearman: **0.8450** · Pearson: **0.4641** —
out-degree is a useful global signal.

Within Top Out-Degree Group

Spearman drops to **0.2133**. Once
structurally strong, **directed position**
matters more than local degree alone.

⚠ Random group shows perfect
correlation — but driven by
near-zero variance, not a
meaningful mechanism.

Representative Cases



BFS level distribution for user 14454 — peak spread at depth 5, reaching 79,859 nodes.

User 14454

Top out-degree group. Reaches **79,859 nodes** — strongest source overall.

User 50329

Out-degree only **26**, yet reaches **74,734 nodes** at max depth 15 — favorable upstream position beats high degree.

Conclusions & Limitations



Key Finding

Directed structural position is more important than early appearance. Out-degree is useful but incomplete.



Limitations

Retweet layer only; unweighted BFS; limited sampled sources; no time-aware cascade reconstruction.



Network Structure

Broadly weakly connected but extremely sparse in the strong sense — confirming a highly directional diffusion structure.

